

EE-Vibe

where you sta
more.



33 Bridges

Karoline Leavitt ,
27, Wears
Probably The
Smallest
The Worl

\$129,900

\$110,000

\$96,

etro Sherry Se

- Cohn & Company

\$120,000

\$247,000

\$89,

4HS Ambassac

Broker Group Richm

Menu

Computing

Mathematics

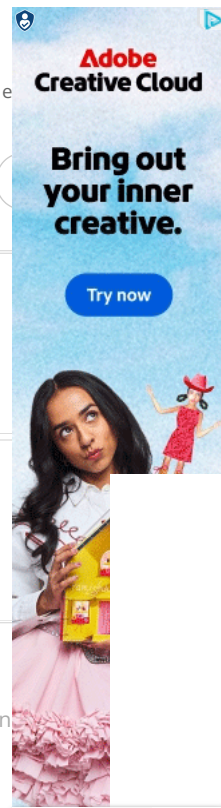
Hardware Devices

Write For Us

Lin

Services

Services



How to Design Counters Using Sequential Circuits?

December 23, 2023 Asma Mushtaq

Search...

Se

Latest

- » Why Adding a Los Angeles Val Parking Service Makes Small Gatherings Feel Grand
- » Electric Boiler : Working, Type: Differences, Maintenance & It Applications
- » Quantum Machine Learning
- » How to Design Half Adder and Adder Circuits?
- » What is Feature Extraction?



Introduction

LifeGr
to be

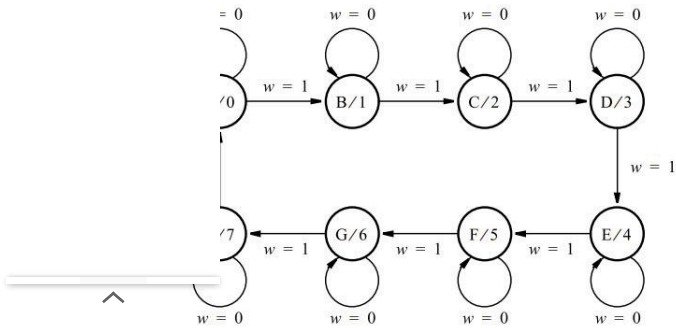
How to Design Counters Using Sequential Circuits? We already know that counters can be implemented like cascaded flip-flop stages and some gate logic where each stage divides a number of incoming pulses by two. To keep our exam size a small counter size, but also show how the design can be extended to larger sizes. Spunter is:

- The counting sequence is 0, 1, ...
- There is an input w

The value of the output is 0 when w = 0, and 1 when w = 1. The counter can be designed as a synchronous sequential circuit using design techniques introduced in the previous section.

State Assignment

Three state variables y_2, y_1 and y_0 . Let us assign binary numbers to encode each output signal in each state. This leads to the table of state assignment.



State Assignment

Present state	Next state		Output
	w = 0	w = 1	
A	A	B	0
B	B	C	1
C	C	D	2
D	D	E	3
E	E	F	4
F	F	G	5
G	G	H	6
H	H	A	7

State Assignment table

After assigning the A,B,...,H to binary numbers.

Present state	Next state		Count
	w = 0	w = 1	
	$y_2 y_1 y_0$	$y_2 y_1 y_0$	$z_2 z_1 z_0$
A	000	001	000
B	001	010	001
C	010	011	010
D	011	100	011
E	100	101	100
F	101	110	101
G	110	111	110
H	111	000	111

state assignment table for counters

The last step in the design is the selection of the type of flip-flops and the derivation of expressions that implement and control the flip inputs. The easiest way to follow this

in \$300
h a
eGreen®
eeking
ount.

se How



- » November 2025
- » October 2025
- » September 2025
- » August 2025
- » July 2025
- » June 2025
- » May 2025
- » April 2025
- » March 2025
- » February 2025
- » January 2025
- » December 2024
- » November 2024
- » September 2024
- » July 2024
- » June 2024
- » May 2024
- » April 2024
- » March 2024
- » February 2024
- » January 2024
- » December 2023
- » June 2023
- » May 2023
- » January 2023
- » December 2022
- » November 2022
- » October 2022
- » September 2022
- » August 2022
- » July 2022

approach. Then we will show the alternative of using JK type flip-flops. In both cases, the flip-flops must be edge-triggered to ensure that it only takes one transition place during one clock cycle.

» June 2022
» May 2022



This
can'

Implementat

When using D-
D input of a flip
information. By
ment a finite state machine, each next-state function Y_i , is connected to
s the specific state variable y_i . the next state functions are derived from tl
: the following results:

$$D_0 = Y_0 = wy_0 +$$

$$D_1 = Y_1 = wy_1 +$$

$$D_2 = Y_2 = wy_2 +$$

It is not clear h
expressions fo

$$D_0 = wy_0 + wy_0$$

$$= w \oplus y_0$$

$$D_1 = wy_1 + y_1 y_0 + wy_0 y_1$$

$$= (w + y_0) y_1 + wy_0 y_1$$

$$= wy_0 y_1 + wy_0 y_1$$

$$= wy_1 \oplus y_1$$

$$D_2 = wy_2 + y_0 y_2 + y_1 y_2 + wy_0 y_1 y_2$$

$$= (w + y_0 + y_1) y_2 + wy_0 y_1 y_2$$

$$= wy_1 y_1 y_2 + wy_0 y_1 y_2$$

$$= wy_0 y_1 \oplus y_2$$

ge implement a larger counter because no clear pattern is discernible in
ever, these expressions can also be written as:

2
:022
22
2021
:2021
2021
r 2021
21

1
:021



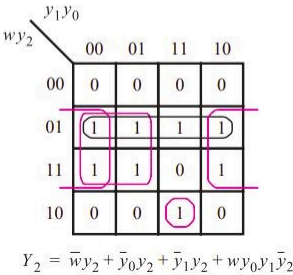
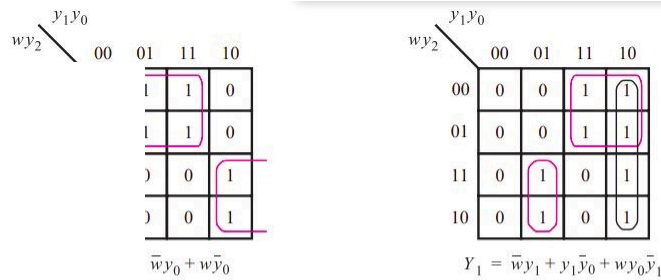
**Remix
your
reality.**

Bring it all to
life with design,
animation,
and more.

Try now



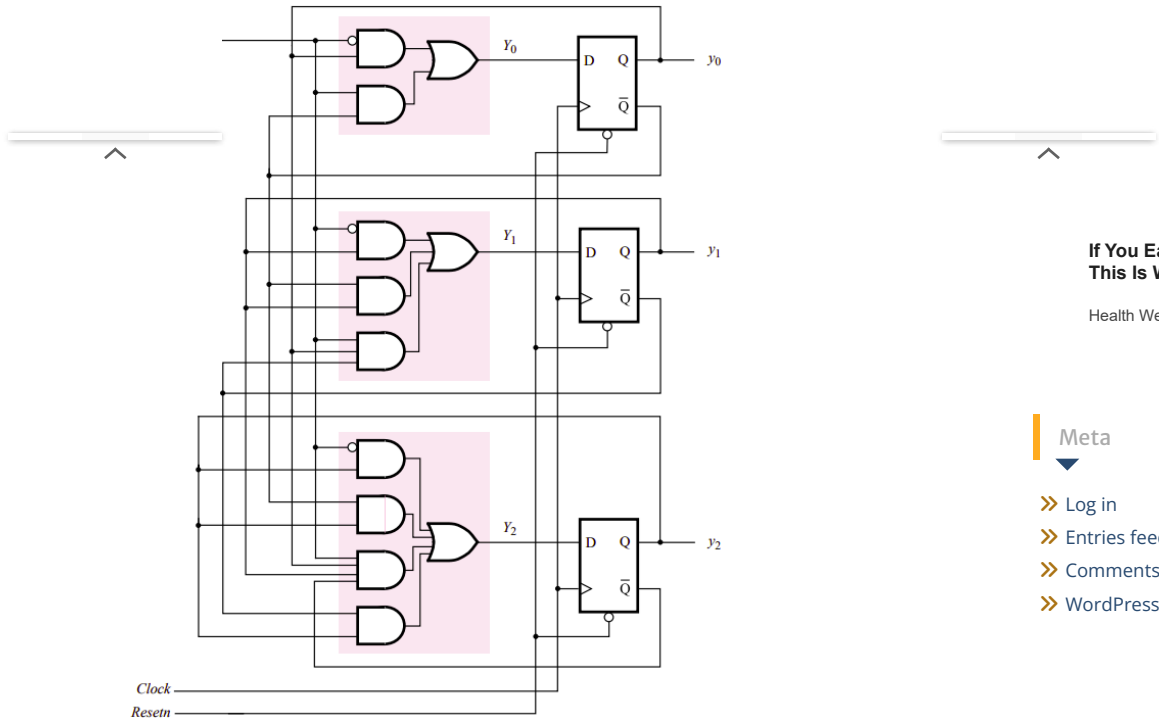
- Categories
- » Arduino Based Projects
 - » Best Final year projects for ele engineering
 - » blog
 - » Computing
 - » CPUs
 - » Devices
 - » Digital Logic Design
 - » education
 - » Electrical Machine
 - » electromagnetic fi ory
 - » Electronic Circuits
 - » Electronics



Karnaugh maps for D flip-flops for the counter.

Circuit diagram

Implemented with D flip-flops



Circuit diagram for the counter implemented with D flip-flops.

Implementation using J-K flipflop:

JK flip flops are an attractive alternative. Using these flip-flops to implement the sequential circuit requires the derivation of the J and K inputs for each flip flop. The following check is required:

- If a flip-flop is in the state 0 it has to remain in the state 0, then the J = 0 and K = d (where d means that K could be equal to either 0 or 1).
- If the flip-flop in state 0 changes its state into 1, then J = 1 and K = d.
- If a flip-flop in state 1 is to remain in state 1, then J = d and K = 0.
- If the flip-flop is in state 1 is to change to state 0, then J = d and K = 1.

Following these instructions, we can create a truth table that specifies the desired values of inputs J and K for the three flip-flops in our design. a modified version of the state-assigned table with input functions J and K included. To see how this table is derived, consider the first row of the state-assigned table where Y2Y1Y0 = 000. If w = 0, then the next state is also Y2Y1Y0 = 000. the value of each flip

- » Elementary Math
- » Engineering Projects

Devices
on to computing
abra
ics

ard
analysis
ented Programming
Microcontroller
Microcontroller

on MPLAB

Pi

If You E:
This Is V
Health We

- Meta
- » Log in
- » Entries feed
- » Comments feed
- » WordPress.org

This means that checking $J = 0$ and $K = d$ for all three flip-flops. We continue in the first line, if $w = 1$, the next state will be $Y_2Y_1Y_0 = 001$. So flip-flops y_2 and y_1 still remain at 0 and have control $J = 0$ and $K = d$. However, flip-flop y_0 must change from 0 to 1. This is achieved with $J = 1$ and $K = d$.



If You Eat Butt

It happens



Lara Is Retiring

Her pieces Are 80% Off

The rest of the table is derived in the same way by considering each gift state $y_2y_1y_0$ and providing the necessary control signals to achieve the new state $Y_2Y_1Y_0$. A state-assigned table is essentially a state table in which each state is encoded state variables.

When D flip-flops are used to implement the FSM, the next state items in the state-assigned table correspond directly to the signals that must be applied to D inputs. This does not apply if another type of flip-flops is used.

The table that gives status information in the form of flip-flop inputs that must be “excited” to cause the transitions to the next states are usually called the excitation table. Wake-up table shows how JK flip-flops can be used.



This
can't

In many books the term excitation table is used even when D flip-flops are involved, in which case it is synonymous with state-assigned table. Once has been derived, it provides a truth table with inputs y_2, y_1, y_0 and w and outputs J_2, K_2, J_1, K_1, J_0 and K_0 . We can then derive expressions for

$$J_0 = K_0 = w$$

$$J_1 = K_1 = wy_0$$

$$J_2 = K_2 = wy_0y_1$$

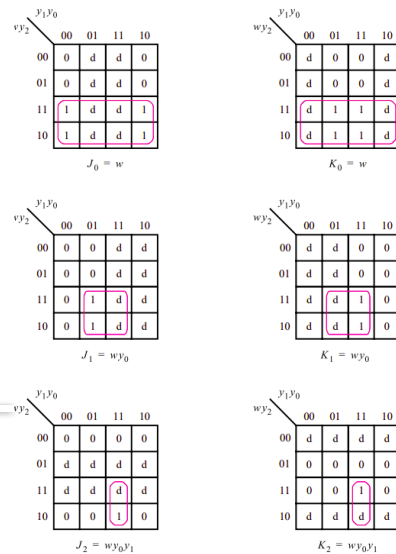


Flip-flop inputs								Count $z_2 z_1 z_0$
$w = 0$				$w = 1$				
$Y_1 Y_0$	$J_2 K_2$	$J_1 K_1$	$J_0 K_0$	$Y_2 Y_1 Y_0$	$J_2 K_2$	$J_1 K_1$	$J_0 K_0$	
000	0d	0d	0d	001	0d	0d	1d	000
001	0d	0d	d0	010	0d	1d	d1	001
010	0d	d0	0d	011	0d	d0	1d	010
011	0d	d0	d0	100	1d	d1	d1	011
100	d0	0d	0d	101	d0	0d	1d	100
101	d0	0d	d0	110	d0	1d	d1	101
110	d0	d0	0d	111	d0	d0	1d	110
111	d0	d0	d0	000	d1	d1	d1	111

Excitation table for the counter with JK flip-flops.

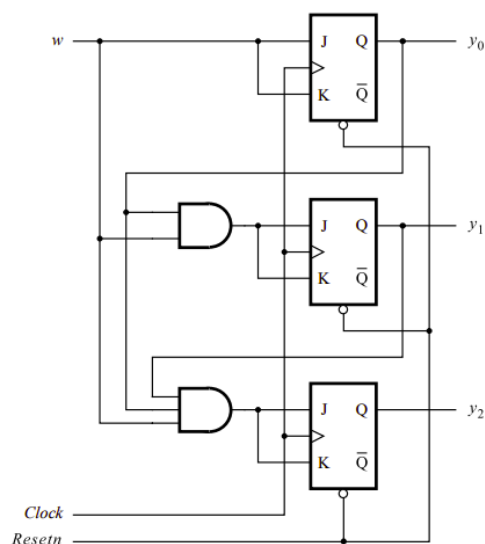
The k-map of t

below:



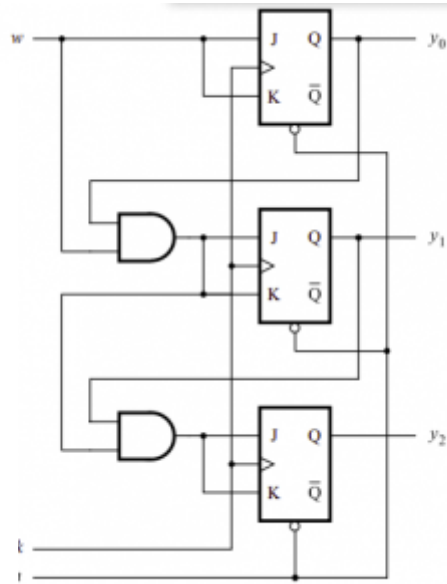
Karnaugh maps for JK flip-flops in the counter.

The logic diagram is given below:



Circuit diagram using JK flip-flops

Using the factored form, the counter circuit can also be implemented. Note that in this circuit all the steps (except the very first one) look the same. Also note that this circuit has the same structure like the previous circuit because it connects the J and K inputs of the flip-flop together they change the flip-flop to a T flip-flop.



Factored-form implementation of the counter.

What are the operations of UP Counter?

What

P Counter Operation



Also read here

- What are the CMOS Logic Gates?
- What is the magnitude comparator circuit? Design a 3 bit magnitude comparator circuit
- What are the synchronous counters? Explain with an example.
- what are the half adder and full adder circuits?
- what are the half subtractor and full subtractor circuits?
- How to design a four bit adder-subtractor circuit?
- What are number systems in computer?
- Discuss the binary counter with parallel load? Explain its working with an example
- how to draw state diagram of sequential circuit?
- How to simplify a Boolean function using Karnaugh map (k-map)?
- What are the flip flops and registers in digital design?
- Define fan-in, fan-out, CMOS and TTL logic levels
- what is the Canonical form representation of Boolean functions?
- What is difference between latches and flip flops?

[An Overview](#)[Why Circuit Theory Fails at High Frequen](#)

Leave a Reply

Your email address

Required fields are marked *

Comment *

Name

Email

Website

☐ Save my name

this browser for the next time I comment.

Post Comment



Copyright. All rights reserved.

EEVIBES Aims to help students by providing education material to them.

